Lingrui Lin, Ph.D. student



I am a Ph.D. student from the School of Astronomy and Space Science of Nanjing University (NJU). I am very easygoing and open-minded, so please feel free to contact me.
[*Last update: October 15, 2026*]

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Education

Sep 2021 – present	**Ph.D. student, School of Astronomy and Space Science, Nanjing University** Thesis: *Multi-scale gas dynamics in galaxies* Expected completion: June 2027 Supervisor: *Zhi-Yu Zhang (NJU)* Research hosts and collaborators: *Federico Lelli (INAF), Carlos De Breuck (ESO)*
Sep 2017 – Jun 2021	**B.Sc., School of Astronomy and Space Science, Nanjing University** Thesis: *Kinematics of atomic gas in supernova remnant IC 443* Supervisor: *Zhi-Yu Zhang*

Research Experience

Sep 8, 2025 – Sep 9, 2026	**Visiting Ph.D. student (funded by CSC),** Arcetri Astrophysical Observatory (INAF), Florence, Italy Topic: *Cold gas dynamics of rotating disk galaxies at* $z = 4$*–5* TRICEPS (Tracing Resolved Ionized Carbon Emission in Primeval Systems): second author of the survey paper (I); first author of the kinematic paper (III), both under review at *Astronomy & Astrophysics* Reduced all ALMA data presented in TRICEPS I and III and all VLA data obtained to date Host: *Federico Lelli*
May 2026	**Scientific visit,** European Southern Observatory (ESO), Garching, Germany Presented a talk at Galaxy Evolution Coffee on May 21, 2026
July 26, 2025 – Aug 2, 2025	**1st China Astrochemistry Summer School,** Xiamen University, Xiamen, China Poster: *Measuring H I/H_2 in Galactic Low-Metallicity Molecular Clouds*; poster prize
July 7, 2025 – July 11, 2025	**JWST Data and Science Summer School,** Westlake University, Hangzhou, China Selected participant and speaker; hands-on training in JWST data reduction and processing
Oct 1, 2023 – Dec 22, 2023	**Early-Career Scientific Visitor,** European Southern Observatory (ESO), Garching bei München, Germany Topic: *[C I] kinematics of a starburst radio galaxy at cosmic noon* Host: *Carlos De Breuck*
Sep 23, 2023 – Sep 30, 2023	**Visitor,** Arcetri Astrophysical Observatory (INAF), Florence, Italy Topic: *[C I] kinematics of a starburst radio galaxy at cosmic noon* Host: *Federico Lelli*

Publications

First-author papers

[1] **L. Lin**, Z.-Y. Zhang, J. Wang, *et al.*, "Inadequate turbulent support in low-metallicity molecular clouds," *Nature Astronomy*, vol. 9, pp. 406–416, Mar. 2025. DOI: 10.1038/s41550-024-02440-3.

[2] **L. Lin**, F. Lelli, C. De Breuck, *et al.*, "Gas dynamics in an AGN-host galaxy at $z \simeq 2.6$: Regular rotation, noncircular motions, and mass models," *Astron. Astrophys.*, vol. 693, A91, Jan. 2025. DOI: 10.1051/0004-6361/202450814.

Coauthored papers

[3] Guo, Wang, **L. Lin**, *et al.*, "Constrain the Size of Broad Line Region in NGC 1068 Using Sub-millimeter Excess," *Mon. Not. R. Astron. Soc.*, Oct. 2026, Accepted.

[4] G. Luo, Z.-Y. Zhang, T. G. Bisbas, *et al.*, "Anomalously high deuterium fractionation in a galactic translucent cloud: A challenge to chemical models," *Astron. Astrophys.*, vol. 712, p. L17, Aug. 2026. DOI: 10.1051/0004-6361/202661031.

[5] Z. Lu, J. Wang, **L. Lin**, *et al.*, "Sharp line profile of radio recombination line H29α in W49A," *Publ. Astron. Soc. Aus.*, vol. 43, e030, Feb. 2026. DOI: 10.1017/pasa.2026.10165.

[6] XSMT Project Collaboration Group, Y. Ao, J. Chang, *et al.*, "Scientific Objectives of the Xue-shan-mu-chang 15-meter Submillimeter Telescope," *Science China Physics, Mechanics & Astronomy*, Sep. 2025, Accepted; arXiv:2509.13983. DOI: 10.48550/arXiv.2509.13983.

[7] W. Wang, C. De Breuck, D. Wylezalek, *et al.*, "JWST + ALMA ubiquitously discover companion systems within $\lesssim$18 kpc around four $z \approx 3.5$ luminous radio-loud AGN," *Astron. Astrophys.*, vol. 696, A88, Apr. 2025. DOI: 10.1051/0004-6361/202553668.

[8] H.-T. Huang, A. W. S. Man, F. Lelli, *et al.*, "Molecular Gas Mass Measurements of an Active, Starburst Galaxy at $z \approx 2.6$ Using ALMA Observations of the [C I], CO, and Dust Emission," *Astrophys. J.*, vol. 977, no. 2, p. 251, Dec. 2024. DOI: 10.3847/1538-4357/ad9021.

[9] G. Luo, D. Li, Z.-Y. Zhang, *et al.*, "The CO-dark molecular gas in the cold H I arc," *Astron. Astrophys.*, vol. 685, p. L12, May 2024. DOI: 10.1051/0004-6361/202450067.

[10] Y. Sun, Z.-Y. Zhang, J. Wang, *et al.*, "An improved method to measure ^{12}C/^{13}C and ^{14}N/^{15}N abundance ratios: revisiting CN isotopologues in the Galactic outer disc," *Mon. Not. R. Astron. Soc.*, vol. 527, no. 3, pp. 8151–8192, Jan. 2024. DOI: 10.1093/mnras/stad3643.

[11] F. Li, Z.-Y. Zhang, J. Wang, G. Luo, **L. Lin**, and J. Zhou, "Dense gas properties around the centre of the Circinus galaxy," *Mon. Not. R. Astron. Soc.*, vol. 527, no. 1, pp. 531–543, Jan. 2024. DOI: 10.1093/mnras/stad3241.

[12] L. Zhang, Z.-Y. Zhang, J. W. Nightingale, *et al.*, "Discovery of a radio jet in the Cloverleaf quasar at z = 2.56," *Mon. Not. R. Astron. Soc.*, vol. 524, no. 3, pp. 3671–3682, Sep. 2023. DOI: 10.1093/mnras/stad2069.

[13] F. Lelli, Z.-Y. Zhang, T. G. Bisbas, *et al.*, "Cold gas disks in main-sequence galaxies at cosmic noon: Low turbulence, flat rotation curves, and disk-halo degeneracy," *Astron. Astrophys.*, vol. 672, A106, Apr. 2023. DOI: 10.1051/0004-6361/202245105.

[14] Y. Deng, Z.-Y. Zhang, P. Zhou, *et al.*, "Multiple gas phases in supernova remnant IC 443: mapping shocked H_2 with VLT/KMOS," *Mon. Not. R. Astron. Soc.*, vol. 518, no. 2, pp. 2320–2340, Jan. 2023. DOI: 10.1093/mnras/stac3139.

[15] G. Luo, Z.-Y. Zhang, T. G. Bisbas, *et al.*, "Dependence of Chemical Abundance on the Cosmic-Ray Ionization Rate in IC 348," *Astrophys. J.*, vol. 942, no. 2, p. 101, Jan. 2023. DOI: 10.3847/1538-4357/aca657.

Under review

[16] **L. Lin**, F. Lelli, C. Bacchini, *et al.*, "TRICEPS. III. Cold gas disks in massive galaxies at $z = 4$–5: fast rotation and low pressure support," Astronomy & Astrophysics, under review, 2026.

[17] F. Lelli, **L. Lin**, Z.-Y. Zhang, *et al.*, "TRICEPS. I. Tracing cold gas, dust, and stars at sub-kpc scales in massive star-forming galaxies at $z = 4$–5," Astronomy & Astrophysics, under review, Sep. 2026. arXiv: 2609.30375 [astro-ph.GA].

[18] Li, Zhang, **L. Lin**, *et al.*, "The outflow in the luminous infrared starburst galaxy IRAS 18293-3413: a multi-phase view," Astronomy & Astrophysics, under review, 2026.

Others

[19] **L. Lin**, Z.-Y. Zhang, and J. Wang, *The initial condition for star formation varies with metallicity: insights from the Galactic outer disk molecular clouds*, Chinese Science Bulletin, 70, 3367–3369, Scientific commentary, in Chinese, 2025. DOI: 10.1360/CSB-2025-0454.

[20] C. De Breuck, B. Emonts, W. Wang, *et al.*, "ALMA observations of high redshift radio galaxies," in *ALMA at 10 years: Past, Present, and Future*, Dec. 2023, p. 50. DOI: 10.5281/zenodo.10244516.

Research Coverage

Feb 2025	INAF / Media INAF, Italy (Italian press coverage of Lin et al. 2025, *Nature Astronomy*)
Jan 2025	Aristotle University of Thessaloniki, Greece (Greek press release)
	Nature Astronomy (English Research Briefing)
	Nanjing University, China (Chinese press release)

Meetings & Conferences

July 10, 2026	Contributed talk, "Dynamically cold gas disks in massive galaxies at $z = 4$–5 from the TRICEPS survey", "How stars move around: what do they tell us about galaxies?", Sexten, Italy
July 3, 2026	ePoster with flash talk, "Dynamics of massive galaxy disks at $z = 4$–5 from the TRICEPS survey", EAS 2026 (SS3a; poster 1817), Lausanne, Switzerland
July 1, 2026	Contributed talk, "Gas and dust in massive galaxies at $z = 4$–5 from the TRICEPS survey", EAS 2026, Lausanne, Switzerland
June 16, 2026	Contributed talk, "Cold gas disks in massive galaxies at $z = 4$–5: fast rotation and low pressure support", Zoom-in Views of Galaxy Disks Across Cosmic Time, Nordita, Stockholm, Sweden
May 21, 2026	Talk, Galaxy Evolution Coffee, ESO Garching
Jan 14, 2026	Talk, Extragalactic Group meeting, INAF–Arcetri
Nov 14, 2024	**Invited Talk,** Annual Conference on Molecular Clouds and Star Formation, Guangxi University, China
Dec 2023	Talk, Galaxy Evolution Coffee, ESO Garching

Observational Projects

As Principal Investigator

ALMA	*An ACA census of Galactic metal-poor molecular clouds* 2021.2.00175.S; ACA total power, Band 6; 64.1 hours
NOEMA	*Resolving the gravity-dominated molecular clouds in the outer Galaxy* W23AY; PolyFiX; 8 hours
SMA	*Galactic metal-poor molecular clouds on sub-pc scales* 2022A-S042; 230/345-GHz receivers, Dual Rx correlator mode; two tracks
IRAM 30-m	*Pebble-sized dust grains in Galactic metal-poor molecular clouds* 105-20; NIKA2; 8.2 hours
JCMT	*Exploring the low dust emissivity index of Galactic metal-poor molecular clouds* M21BP041; SCUBA-2; 0.5 hours
SMT	*Dynamic states of molecular clouds in the outer Galaxy* Lin_L_22B_1; 1.3-mm receiver; 48 hours
FAST	*Measuring H I Fraction in Galactic Metal-poor Molecular Clouds* PT2021_0055; 19-beam receiver/Spec(W+N); 17 hours
	A HINSA census towards metal-poor molecular clouds in the Galactic far outer disk PT2024_0068; 19-beam receiver/Spec(W+N); 17.5 hours
GBT 100-m	*Resolving the turbulence-deficient molecular clouds in the outer Galaxy* GBT24B-167; Argus/VEGAS; 21 hours

Observational Projects (continued)

Effelsberg — *CO-dark molecular gas in the outer Galaxy*
98-22; P180mm/XFFTS; 46.2 hours

As an active collaborator (selected)

ALMA — *Cold gas and dark matter in massive galaxies at $z = 4$–5*
CoI, 2023.1.01724.S (5 hours) and 2025.1.00827.S (47.6 hours); PI Federico Lelli
Ultradeep [C II] observations of rotating gas disks at $z = 4.5$
CoI, 2023.1.00835.S, 2024.1.00839.S, 2025.1.01398.S, and 2026.1.00563.S; PI Federico Lelli
2026.1.00563.S: Band 7, 39.9 hours

VLA — *Cold gas and dark matter in massive galaxies at $z = 4$–5*
CoI, 24A-296 and 25A-307; WIDAR; 101.5 and 109.9 hours respectively; PI Federico Lelli

JWST — *Luminous and dark matter in massive galaxies at $z = 4$–5*
CoI, Cycle 2, program 3954; NIRCam and MIRI; 24.92 hours; PI Federico Lelli

IRAM 30-m — *Dynamic states of molecular clouds in the outer Galaxy*
Lead proposal contributor, 102-22 (2022); PI Thomas Bisbas

JCMT — *Dust in shocked regions of Supernova remnant IC 443*
CoI and lead proposal contributor, M22AF002 (2022); SCUBA-2; 5 hours
Sub-virial molecular clouds in the outer Milky Way on core to parsec scales: Short-spacing for the SMA
CoI and lead proposal contributor, M22BP027; HARP and $\bar{U}\bar{u}$ receivers/ACSIS; 25 hours; PI Hauyu Baobab Liu

FAST — *Atomic Clouds Driven by the Galactic Nuclear Wind*
PT2024_0083; PI Jing Zhou

GBT 100-m — *An OH and CH survey of the molecular clouds in the outer-disk of the Milky Way*
GBT23A-246; PI Gan Luo
Mapping the CO-dark molecular gas in the outer Galaxy (2023)
Lead proposal contributor

Awards & Grants

2026 – present — Zhujian Program for Doctoral Students, Nanjing University
CNY 100,000 (approximately EUR 13,300)

Sep 8, 2025 – Sep 9, 2026 — China Scholarship Council support for the research visit to INAF–Arcetri
EUR 16,200 total

2025 — National Scholarship (China)
Poster prize, 1st China Astrochemistry Summer School, Xiamen University

2021 — Linqiao Scholarship (Linbridge Fund, Douglas Nelson Chao Lin)
Outstanding Undergraduate Graduate Award
President's Special Scholarship for Doctoral Students

2018 – 2020 — Top Talent Program Scholarship $\times$ 3

2019 — Annual Scholarship of National Astronomical Observatory, Chinese Academy of Sciences
Renmin Scholarship

2018 — National Encouragement Scholarship
National College Mathematics Competition (Second prize, Non-Mathematics Major)

Responsibilities

Fall 2026	Organizer, “Discussions on the frontiers of high-z galaxies and the ISM”, Nanjing University
Spring 2025	Teaching Assistant of "Astronomical Literature — Reading and Writing", NJU Astronomy
Spring 2024	Teaching Assistant of "Interstellar Medium", NJU Astronomy
2022 – 2023	Organization Member, Graduate Student Committee of NJU Astronomy
2018 – 2021	Monitor, NJU Astronomy Class 2017
2018 – 2019	Leader, Academic Division of NJU Great Astronomy Fans Association (NJU GAFA)
	Captain, NJU Astronomy Table Tennis team
2017 – 2018	Publicity Department Member, Student Union of NJU Astronomy
	Representative for Studies, NJU Astronomy Class 2017

Skills

Astronomical Expertise

Radio astronomy	Observing preparation; remote observations with SMT, GBT, and Effelsberg
	Spectral-line and continuum data reduction for ALMA, VLA, NOEMA, SMA, IRAM 30-m, JCMT, SMT, GBT, FAST, and Effelsberg
	Calibration, continuum subtraction, and spectral-line and continuum imaging using CASA
Gas dynamics	Three-dimensional disk fitting, rotation-curve mass modeling, and analysis of spectral cubes and velocity fields
Radio astronomy	Combining single-dish and interferometric data in the uv plane
Astronomical Tools	*Proficient usage:* GILDAS/CLASS, GILDAS/Mapping, CASA, Carta, DS9, ^{3D}Barolo, `vcdisk`, Astropy, Photutils, Montage
	Basic usage: GILDAS/PIIC, Starlink, MIR, Glue
Code development	**See more on GitHub: Astronlin**
	Highlight: (a) A Python FAST H I calibration pipeline; *(b)* MCMC-based HINSA fitting tools; *(c)* interactive scientific visualizations

Technical Skills

Coding	Python (proficient), Fortran (basic), C++ (basic), IDL (basic)
Operating System	Linux (I help maintain the Ubuntu/CentOS servers and remote-access services in the NJU group), macOS, Windows
Writing and editing	Vim (proficient), LaTeX (proficient), Microsoft Word (proficient)

Miscellaneous

Languages	Mandarin Chinese (Native), English (Fluent)
Sports	Table tennis (ping-pong; advanced), Swimming (proficient), Basketball (competent), Badminton (competent), Tennis (novice)
Photography	Adobe Photoshop/Camera Raw, Adobe Premiere